

Literature status: the unbounded Davies–Gaffney–Grigor’yan question for Hodge Laplacians

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2026-07-18

Status. This packet records a `literature_already_answered` identification. It is not an original proof from the run.

Original question. Bobo Hua and Xin Luo, *Davies–Gaffney–Grigor’yan lemma on simplicial complexes*, arXiv:1702.00529, prove a Davies–Gaffney–Grigor’yan estimate for heat kernels of bounded discrete Hodge Laplacians on simplicial complexes. Their Remark 1.1 on page 3 says: “It will be interesting to know whether it holds for unbounded cases.” Here “it” is Theorem 1.2, the DGG lemma for bounded Hodge Laplacians.

Answering paper. Philipp Bartmann and Matthias Keller, *The heat equation and independence of the spectrum of the Hodge Laplacian on ℓ^p* , arXiv:2602.18597, explicitly answers this question. The introduction says that the paper establishes a DGG estimate for the heat semigroup kernel and that this answers the question in Hua–Luo concerning unbounded operators. In Section 5.2, immediately before Theorem 5.1, the authors state that their weighted-simplicial-complex DGG estimate answers Hua–Luo Remark 1.1.

Why this answers the question. Bartmann–Keller first prove Theorem 2.2, a DGG estimate for general positive magnetic Schrodinger operators on graphs associated with a positive closable quadratic form and an intrinsic metric of finite jump size. They then use the signed-Schrodinger representation of discrete Hodge Laplacians on weighted simplicial complexes. The resulting Theorem 5.1 gives, for each $\circ \in \{+, -, H\}$ and all $\omega, \eta \in \ell^2$,

$$|\langle e^{-t\Delta_2^\circ} \omega, \eta \rangle| \leq \exp\left(-\lambda_2^\circ t - \frac{t}{s^2} \zeta\left(\frac{s d^\circ(\text{supp } \omega, \text{supp } \eta)}{t}\right)\right) \|\omega\|_2 \|\eta\|_2,$$

with the corresponding pointwise heat-kernel bound. The theorem is stated for the unbounded Dirichlet Hodge Laplacians arising from the closed forms, so it removes exactly the boundedness restriction singled out in Hua–Luo Remark 1.1.

Scope. The answer covers the DGG estimate for the unbounded Hodge Laplacians in the weighted simplicial-complex framework used by Bartmann–Keller, via their magnetic Schrodinger operator reduction. This packet makes no separate claim about other open directions in ℓ^p Hodge theory.

Evidence used. The source question was checked against arXiv:1702.00529 and the supporting answer against arXiv:2602.18597. Cheap run indexes had no previous record for either arXiv id at the time of packaging.

References

- [1] B. Hua and X. Luo, *Davies–Gaffney–Grigor’yan lemma on simplicial complexes*, arXiv:1702.00529; Math. Z. 290 (2018), 1041–1053.
- [2] P. Bartmann and M. Keller, *The heat equation and independence of the spectrum of the Hodge Laplacian on ℓ^p* , arXiv:2602.18597.